THE PROBLEM

Money arrives. Spending it shouldn't need a cash trip.

- Remittances often become cash at a pickup counter before they can buy food or medicine.
- Many recipients have phones but patchy or expensive mobile data.
- Senders can't easily earmark support for everyday essentials.
- Neighbourhood shops extend informal credit and carry the risk.

A household story

Maria works in Toronto and supports her mother Rosa in Oaxaca. Rosa buys groceries at the shop down her street. With OpenTab, Rosa scans the shop's QR, approves on her phone with no data, and the shop is paid on Stellar seconds later. No pickup counter, no cash to carry.

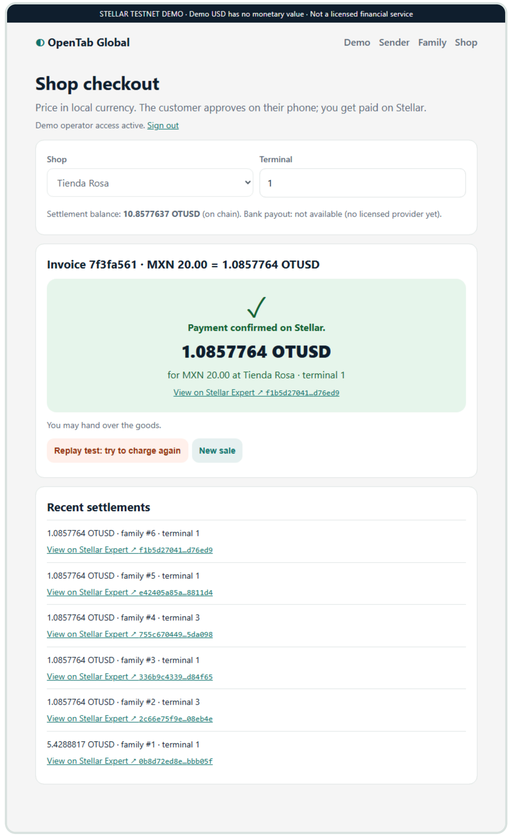
Illustrative household story. We have not yet run user interviews or pilots.

PROBLEM SPACE 3 · TRANSNATIONAL HOUSEHOLDS

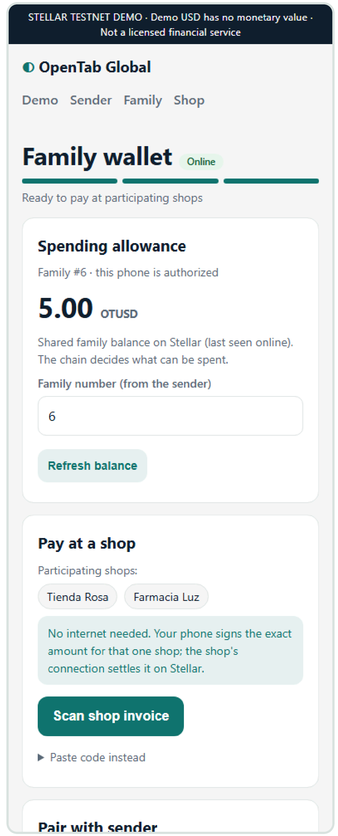
OpenTab Global

A shared family spending balance on Stellar, spendable at local shops, approved on a phone **with no internet**.

SENDER · ONLINE **Funds** the family balance → SHOP · ONLINE **Invoice QR** local price + exact charge → PHONE · OFFLINE **Signs** that exact purchase → SHOP · ONLINE **Submits** the approval → STELLAR **Pays the shop** once; replays refused



Shop: "Payment confirmed" only after Stellar confirms (real Testnet tx)



Family phone: allowance read from the contract



Offline: "Authorized. Awaiting merchant settlement."

Screenshots are from the working app on Stellar Testnet. Demo token, illustrative exchange rate.

BUILT ON STELLAR

Rules enforced by a smart contract

- **FamilyVault** (Soroban): family balances, phone keys, one-time invoices.
- Phone signs 340 bytes binding network, contract, token, family, shop, terminal, invoice, amount, price, expiry.
- Contract verifies Ed25519 natively: the family member needs **no account and no XLM**.
- Stellar Asset Contract holds and pays the demo dollar token.

Verified on Testnet

- Contract `CAUCH3MN…ABYLGW` deployed
- Payments at 2 shops from 1 balance
- Replayed approval refused; overspend refused
- 34 contract · 13 protocol · 18 API · 18 security tests
- Access-controlled demo; secrets never in the browser

Live Connected shop, offline customer

In development Both offline: funds reserved for one named shop, running-total tickets, settle later

Roadmap

- Physical phone pilots
- Real USDC
- Licensed local payout partners
- Audit

Testnet prototype · OTUSD demo token has no value and is not USDC · exchange rates illustrative · no bank payouts · unaudited.